

DNS SERVER

(BIND9)

NAMA : BRILIAN ABEDNEGO WIRYAWAN
KELAS : XI TJKT 1
ABSEN : 6

1. Hal yang dilakukan pertama adalah **login akun root su -**

```
brilian@brilian:~$ su -  
Password:  
root@brilian:~# pwd
```

2. Selanjutnya lakukan setting ip untuk client (statis) dan dan internet (dynamic) **NANO**
/ETC/NETWORK/INTERFACES.

```
root@brilian:~# nano /etc/network/interfaces
```

3. Sesuaikan coding seperti gambar di bawah:

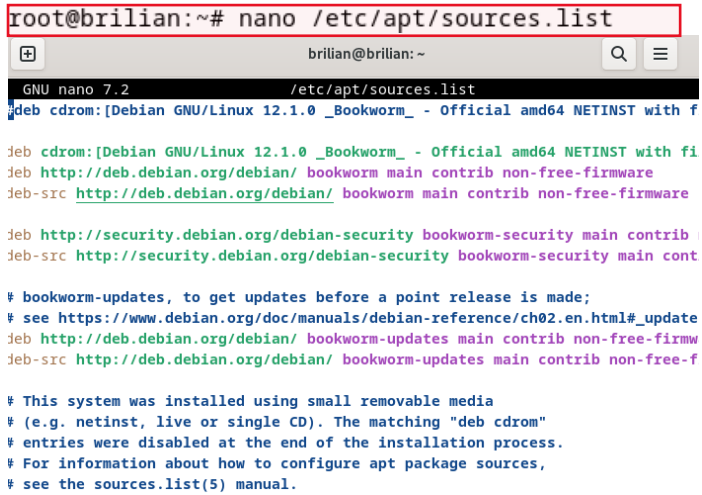
```
GNU nano 7.2 /etc/network/interfaces *  
# This file describes the network interfaces available on your system  
# and how to activate them. For more information, see interfaces(5).  
  
source /etc/network/interfaces.d/*  
  
# The loopback network interface  
auto lo  
iface lo inet loopback  
# enp0s3  
auto enp0s3  
iface enp0s3 inet dhcp  
  
#bakwan goreng  
allow-hotplug enp0s8  
iface enp0s8 inet static  
    address 10.10.6.56  
    netmask 255.255.255.224
```

4. Setelah selesai lakukan restart supaya coding yang dimasukkan dapat diperbarui/di proses.

```
root@brilian:~# /etc/init.d/networking restart  
Restarting networking (via systemctl): networking.service.
```

5. Setelah itu lakukan perintah **SESUAI GAMBAR DIBAWAH**

```
root@brilian:~# nano /etc/apt/sources.list
```



```
GNU nano 7.2 /etc/apt/sources.list
deb cdrom:[Debian GNU/Linux 12.1.0 _Bookworm_ - Official amd64 NETINST with f
deb cdrom:[Debian GNU/Linux 12.1.0 _Bookworm_ - Official amd64 NETINST with fi
deb http://deb.debian.org/debian/ bookworm main contrib non-free-firmware
deb-src http://deb.debian.org/debian/ bookworm main contrib non-free-firmware
deb http://security.debian.org/debian-security bookworm-security main contrib
deb-src http://security.debian.org/debian-security bookworm-security main cont
# bookworm-updates, to get updates before a point release is made;
# see https://www.debian.org/doc/manuals/debian-reference/ch02.en.html#_update
deb http://deb.debian.org/debian/ bookworm-updates main contrib non-free-firmw
deb-src http://deb.debian.org/debian/ bookworm-updates main contrib non-free-f
# This system was installed using small removable media
# (e.g. netinst, live or single CD). The matching "deb cdrom"
# entries were disabled at the end of the installation process.
# For information about how to configure apt package sources,
# see the sources.list(5) manual.
```

6. Lalu lakukan ping untuk mengecek saja.

```
root@brilian:~# ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=115 time=17.5 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=115 time=17.0 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=115 time=17.0 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=115 time=17.0 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=115 time=17.1 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=115 time=17.1 ms
```

7. Lalu, **upgrade dan update sesuai perintah di bawah.**

```
root@brilian:~# apt upgrade&&update
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
```

8. Jika sudah akan seperti gambar dibawah.

```
Fetchd 282 kB in 1s (276 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

9. Jika sudah, lakukan **perintah apt install BIND9** dan jika ada keterangan ingin melanjutkan atau tidak seperti gambar di bawah **tekan Y saja**.

```
root@brilian:~# apt install bind9 dnsutils
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
After this operation, 1,582 kB
Do you want to continue? [Y/n]
```

10. Jika berhasil akan seperti ini.

```
Created symlink /etc/systemd/system/multi-user.target
/systemd/system/named.service.
Processing triggers for man-db (2.11.2-2) ...
root@brilian:~#
```

11. Proses instalasi **BIND9** sudah berhasil dan selesai dilakukan. Selanjutnya kita masuk **directory bind9** dengan perintah.

```
root@brilian:~# cd /etc/bind
```

12. Selanjutnya kita coba check isi **file bind**, gunakan perintah seperti dibawah.

```
root@brilian:/etc/bind# ls
bind.keys  db.255    named.conf      named.conf.options
db.0       db.empty  named.conf.default-zones  rndc.key
db.127     db.local  named.conf.local  zones.rfc1918
```

13. Selanjutnya kita lakukan **copy file** yang ada diatas yang sudah kita munculkan **db. local** dan menambahkan nama dan db.127 dan menambahkan octet pertama ip kita kita dengan **perintah dibawah**. Jika berhasil akan muncul seperti gambar dibawah.

```
root@brilian:/etc/bind# cp db.local db.brilian
root@brilian:/etc/bind# cp db.127 db.10
```

14. Kemudian kita **edit file** yang baru saja kita **copy file**. Ikuti perintah dibawah.

```
root@brilian:/etc/bind# nano db.brilian
```

15. Selanjutnya kita **ubah LocalHost dengan nama kalian dan Menambahkan ip sesuai kebutuhan** kalian.

```
$TTL      604800
@         IN      SOA      brilian.net. root.brilian.net. (
                                2          ; Serial
                                604800     ; Refresh
                                86400      ; Retry
                                2419200    ; Expire
                                604800 )   ; Negative Cache TTL
;
@         IN      NS       brilian.net.
@         IN      A        10.10.6.56
www       IN      A        10.10.6.56
```

16. Kemudian kita **edit file** yang baru saja kita **copy file**. Ikuti perintah dibawah.

```
root@brilian:/etc/bind# nano db.10
```

17. Selanjutnya kita **ubah LocalHost dengan nama kalian dan Menambahkan ip sesuai kebutuhan**.

```

$TTL      604800
@         IN      SOA      brilian.net. root.brilian.net. (
                                1          ; Serial
                                604800     ; Refresh
                                86400      ; Retry
                                2419200    ; Expire
                                604800 )   ; Negative Cache
;
@         IN      NS       brilian.net.
56        IN      PTR      brilian.net.

```

18. Kemudian kita **edit file named.conf.default-zones**. Ikuti perintah dibawah.

```
root@brilian:/etc/bind# nano named.conf.default-zones
```

19. Selanjutnya kita menambahkan **ip dan LocalHost kita**. Ikuti perintah pada gambar dibawah.

```

zone "brilian.net" {
    type master;
    file "/etc/bind/db.brilian";
};

zone "10.10.6.in-addr.arpa" {
    type master;
    file "/etc/bind/db.10";
};

```

20. Kemudian kita **edit file resolv.conf**. Ikuti perintah dibawah.

```
root@brilian:/etc/bind# nano /etc/resolv.conf
```

21. Kemudian **isi NameServer** dengan kebutuhan sendiri. ikuti gambar dibawah.

```

nameserver 10.10.6.56
search brilian.net

```

22. Jika sudah selesai lakukan restart **pada gambar dibawah**.

```
root@brilian:/etc/bind# systemctl restart bind9
```

23. Sekarang Jika sudah selesai lakukan status ikuti pada gambar dibawah.

```

Active: active (running) since Wed 2025-02-26 08:47:33 WIB; 5s ago
Docs: man:named(8)
Main PID: 8769 (named)
Status: "running"
Tasks: 4 (limit: 3987)
Memory: 11.2M
CPU: 40ms
CGroup: /system.slice/named.service
└─8769 /usr/sbin/named -f -u bind

Feb 26 08:47:34 brilian named[8769]: FORMERR resolving './NS/IN': 192.112.36.4#53
Feb 26 08:47:34 brilian named[8769]: DNS format error from 170.247.170.2#53 resolving './NS/IN': 170.247.170.2#53
Feb 26 08:47:34 brilian named[8769]: FORMERR resolving './NS/IN': 170.247.170.2#53
Feb 26 08:47:34 brilian named[8769]: DNS format error from 192.36.148.17#53 resolving './NS/IN': 192.36.148.17#53
Feb 26 08:47:34 brilian named[8769]: FORMERR resolving './NS/IN': 192.36.148.17#53
Feb 26 08:47:34 brilian named[8769]: DNS format error from 198.41.0.4#53 resolving './NS/IN': 198.41.0.4#53
Feb 26 08:47:34 brilian named[8769]: FORMERR resolving './NS/IN': 198.41.0.4#53
Feb 26 08:47:34 brilian named[8769]: DNS format error from 192.33.4.12#53 resolving './NS/IN': 192.33.4.12#53
Feb 26 08:47:34 brilian named[8769]: FORMERR resolving './NS/IN': 192.33.4.12#53
Feb 26 08:47:34 brilian named[8769]: resolver priming query complete: failure
lines 1-22/22 (END)
^C
root@brilian:/etc/bind# systemctl restart bind9
root@brilian:/etc/bind#

```

24. Jika sudah aktif/running [kata yang berwarna hijau] maka sudah berhasil.

25. Kemudian kita lakukan perintah lookup untuk check ip. Ikuti perintah dibawah.

```

root@brilian:~# nslookup 10.10.6.56
56.6.10.10.in-addr.arpa name = brilian.net.

```

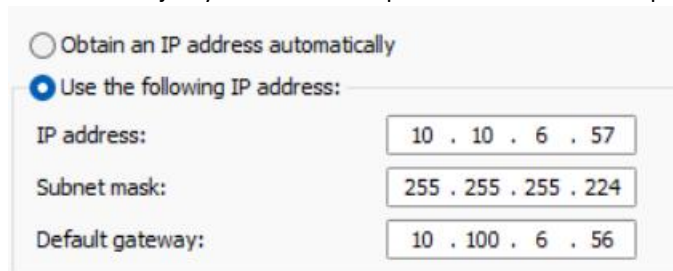
26. Kemudian kita lakukan perintah lookup untuk check DomainName. Ikuti perintah dibawah.

```

root@brilian:~# nslookup brilian.net
Server:          10.10.6.56
Address:         10.10.6.56#53

```

27. selanjutnya kita sediakan ip untuk client di control panel.



The screenshot shows a network configuration window with two radio buttons at the top: "Obtain an IP address automatically" (unselected) and "Use the following IP address:" (selected). Below the radio buttons, there are three input fields with their respective values:

IP address:	10 . 10 . 6 . 57
Subnet mask:	255 . 255 . 255 . 224
Default gateway:	10 . 100 . 6 . 56

28. selanjutnya kita ping ip kita di commant prompt. Lakukan ping dan lookup pada ip dan domain name.

```
C:\Users\ABED NEGO>ping 10.10.6.56

Pinging 10.10.6.56 with 32 bytes of data:
Reply from 10.10.6.56: bytes=32 time<1ms TTL=64
Reply from 10.10.6.56: bytes=32 time<1ms TTL=64
Reply from 10.10.6.56: bytes=32 time<1ms TTL=64
Reply from 10.10.6.56: bytes=32 time<1ms TTL=64

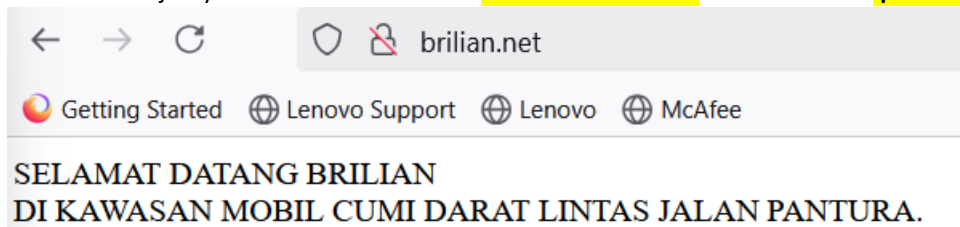
Ping statistics for 10.10.6.56:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\ABED NEGO>nslookup 10.10.6.56
Server:  brilian.net
Address:  10.10.6.56

Name:    brilian.net
Address: 10.10.6.56

C:\Users\ABED NEGO>nslookup brilian.net
Server:  brilian.net
Address: 10.10.6.56
```

29. selanjutnya kita buka browser bisa **chrome atau firefox**. Kita masukkan **Ip atau DomainName**.



SEKIAN TERIMAKASIH.

